Élan

Emergency Cardiac Alert App — Project Report

A plain-language guide to what the app does, who it is for, how it works, and how it keeps people safe. No technical background needed.

Important: Élan helps people reach nearby hospitals faster during a heart emergency. It does **not** replace calling 112 (or 108 for an ambulance).

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APP VERSION
1.0.0 (Android)

SUMMARY

Élan at a glance

Élan is a mobile app that lets a person in a cardiac emergency alert the nearest hospitals with one long press. The hospitals instantly see who needs help, where they are and what they are feeling. When a hospital accepts, the person sees that help is coming and roughly when it will arrive.

1 press
Hold the SOS button for 1.5 seconds to send an alert

3
nearest hospitals are alerted at the same time

₹0
Sending an alert is free; no payment step

2 sides
One app for patients and for hospital duty staff

The name **Élan** (said *ay-LAHN*) is French for "spirit" or "momentum". It reflects the idea at the heart of the app: in a heart emergency, every minute counts, so the app is built to remove delay.

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SECTION 1

The problem and the idea

The problem

During a heart attack, the time between the first symptoms and proper treatment matters a great deal. Yet in that moment, people often lose precious minutes:

- They are not sure which hospital is closest, or which one can take them.
- They phone hospitals one by one, repeating the same details each time.
- The hospital usually learns about the patient only when the patient walks through the door, with no time to prepare.
- While waiting, the patient has no idea whether anyone is coming.

The idea

Élan turns this around. With a single long press, the app sends the person's **exact location**, their **symptoms** and an optional **short message** to the nearest hospitals **all at once**. Hospital staff see the alert on their own screen, can accept it with one tap, and tell the patient how many minutes until their team arrives. The patient watches all of this happen live.

> **In one sentence:** Élan is a "panic button" for heart emergencies that connects a patient directly to the duty desks of nearby hospitals, and shows the patient that help is on the way.

Who uses Élan

Members

Patients, family members and carers. They create an account, and in an emergency they open the app, pick symptoms and hold the SOS button.

They can also keep simple health records: patient details, vital signs (heart rate, blood pressure, temperature, oxygen level) and medical reports.

Hospital staff

Duty staff at a hospital's emergency desk. They sign in on the "Hospital staff" tab and land on a *triage desk*: a live list of incoming alerts.

They accept an alert and choose an arrival time, or pass it to other hospitals. New staff accounts must be approved by an administrator first.

SECTION 2

How an alert works, step by step

Here is the journey of one alert, from the moment someone feels unwell to the moment help is on the way.

1

Open the Emergency screen

From the home screen, the patient taps the big red "Send an emergency alert" card. The app finds their location using the phone's GPS.

Patient

2

See nearby hospitals

The app lists hospitals within about 15 km, closest first, with their distance and phone number.

Patient

3

Pick symptoms and hold to send

The patient taps what they feel (for example "Chest Pain") and can type a short message. Then they press and hold the SOS button for 1.5 seconds. Holding, rather than tapping, prevents accidental alerts.

Patient

4

The 3 nearest hospitals are alerted

Élan's server checks that hospitals are in range and delivers the alert to the three closest ones at the same time. It is free; there is no payment step.

Élan server

5

A hospital accepts

The alert appears on each hospital's triage desk with the patient's name, symptoms, distance and a waiting timer. A staff member taps "5", "10", "15" or "20 min" to accept and say when their team will arrive.

Hospital

6

The patient sees "Help is on the way"

The patient's screen updates by itself: which hospital accepted, a countdown to arrival, and one-tap buttons to call that hospital or 112.

Patient

If something goes wrong, the app says so. If there is no GPS signal, no internet, or no hospital in range, Élan stops and tells the patient clearly that the alert was *not* sent. It never guesses a location or pretends an alert went through. The 112 button is always one tap away.

SECTION 3 · APP TOUR

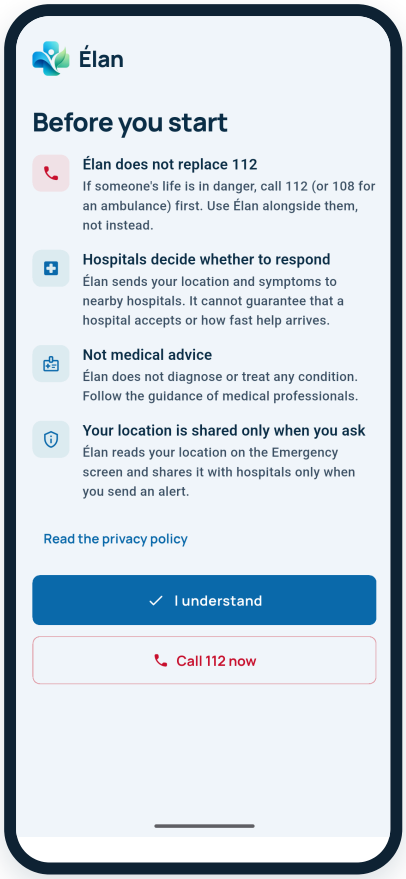
Getting started

All screenshots in this tour are from the real app running on an Android phone emulator, using test accounts and test hospitals.



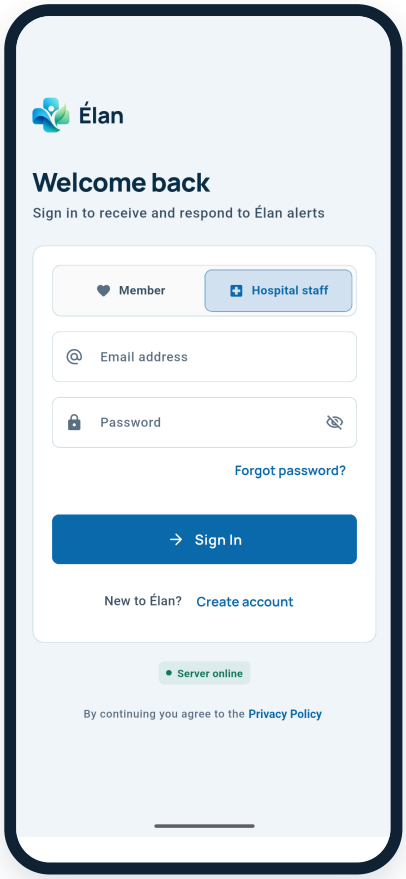
Launch screen

A clean white screen with the Élan logo. It lasts about a second; in an emergency nobody should wait on a logo.



Safety notice (first launch)

Before anything else, the app explains in plain words what it can and cannot do, and offers a "Call 112 now" button.



Sign in

One screen for both kinds of user: "Member" or "Hospital staff". There is a "Forgot password?" link and a live "Server online" indicator.

Why a safety notice first?

People must understand that Élan works **alongside** emergency services, not instead of them. The notice appears once, before the very first sign-in, and is required reading for a health app on Google Play.

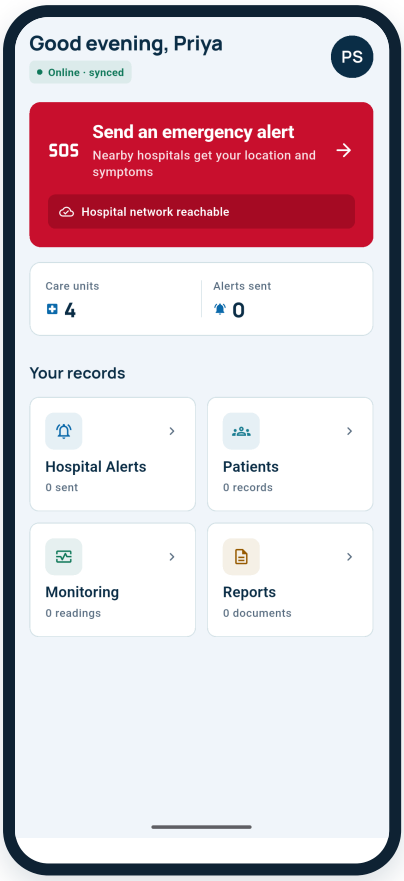
Two kinds of account

The same app serves patients and hospitals. Choosing "Hospital staff" at sign-in opens the hospital's triage desk instead of the patient home screen.

SECTION 3 · APP TOUR

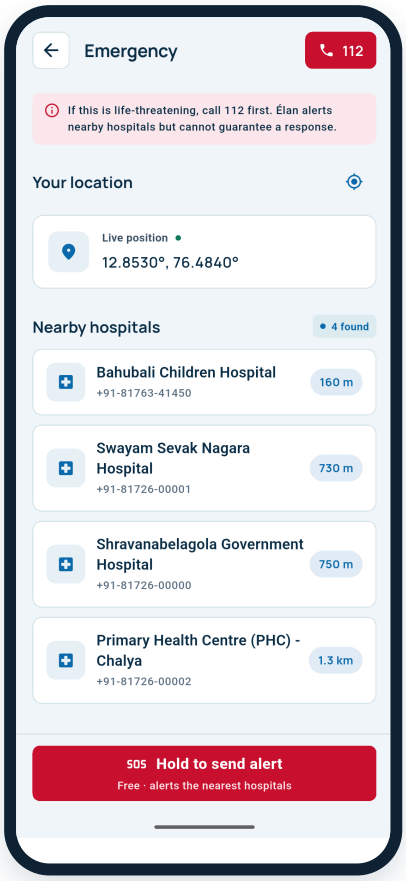
Home and the Emergency screen

The home screen puts the emergency button first. Everything else, such as records, comes after.



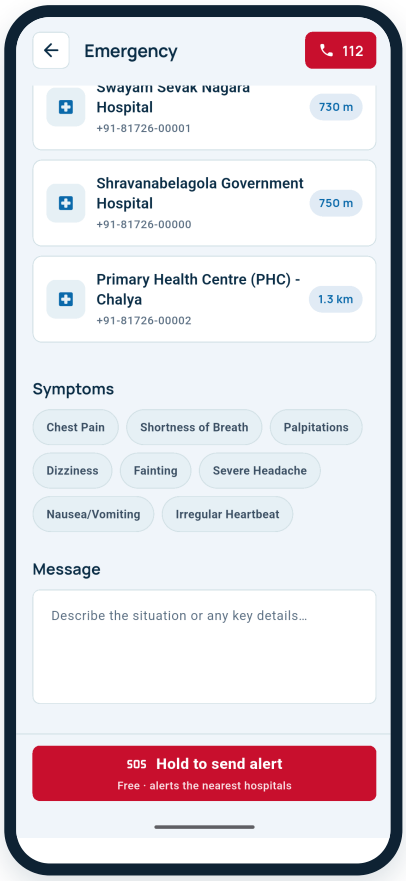
Home

A greeting, a connection status badge, the red SOS card, and shortcuts to alerts, patients, vital-sign monitoring and reports.



Emergency

A red 112 button at the top, a reminder to call 112 first, the patient's live position, and nearby hospitals sorted by distance.



Symptoms and send

Tap symptoms, add a message if able, then press and hold "Hold to send alert". Letting go early cancels.

Why hold instead of tap?

A tap can happen by accident: in a pocket, or by a child. Holding for 1.5 seconds is quick for someone who means it, but very hard to do by mistake. The button fills up as it is held, and letting go early cancels.

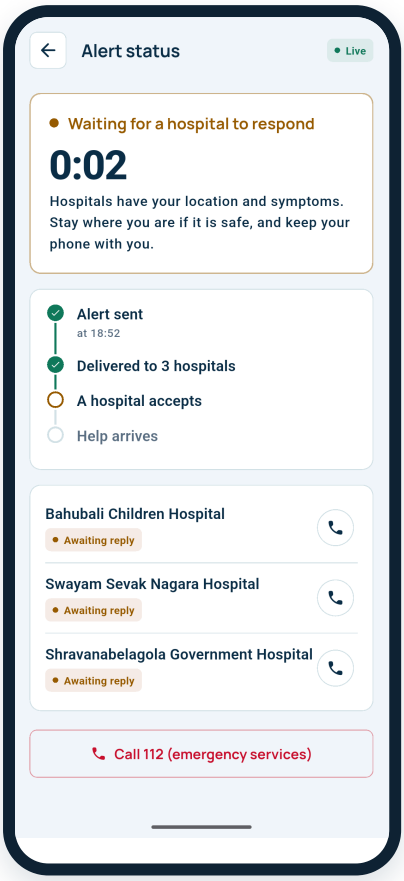
What if no hospital is nearby?

The Emergency screen shows "No hospitals found nearby" and the alert is not sent, so nobody waits for a reply that cannot come. The 112 button stays at the top of the screen.

SECTION 3 · APP TOUR

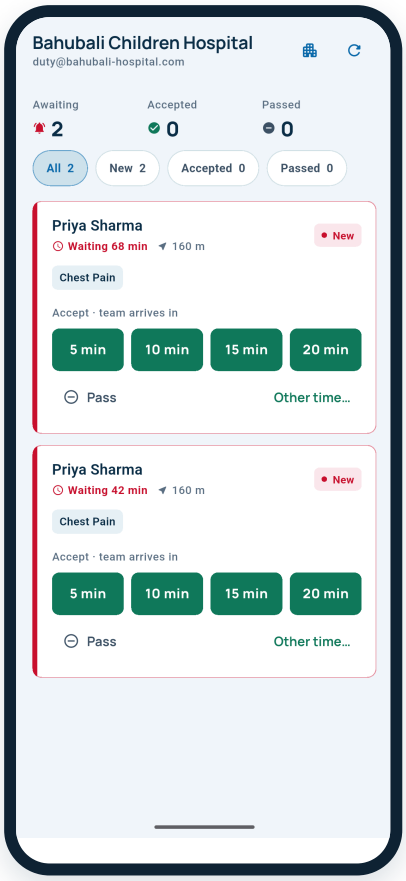
From alert to "help is on the way"

This is the same alert seen from both sides: the patient's phone and the hospital's duty desk.



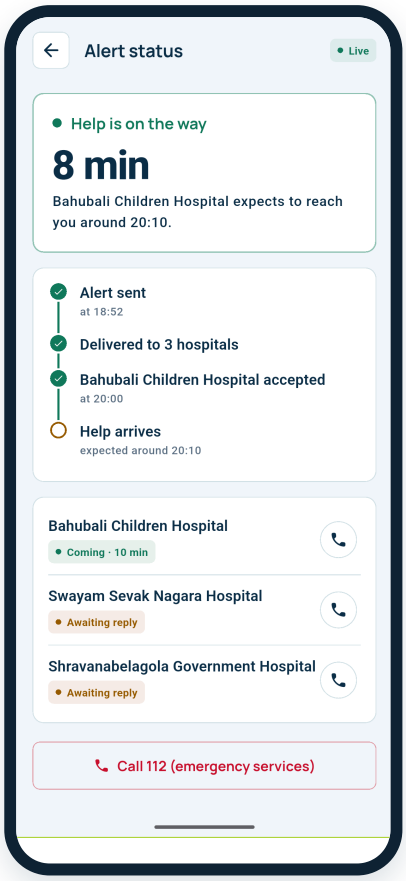
Patient: alert sent

A running timer and a simple timeline: sent, delivered to 3 hospitals, waiting for one to accept. Each hospital can be called with one tap.



Hospital: triage desk

Unanswered alerts are listed first. The waiting time turns red after 2 minutes. Staff accept with an arrival time, or pass.



Patient: help is coming

Once a hospital accepts, the patient sees which hospital, a countdown to arrival, and the expected time.

What the hospital sees

The patient's name, symptoms, message, distance and exact location, plus how long the patient has been waiting. Staff choose an arrival time, or "Other time…" for anything else.

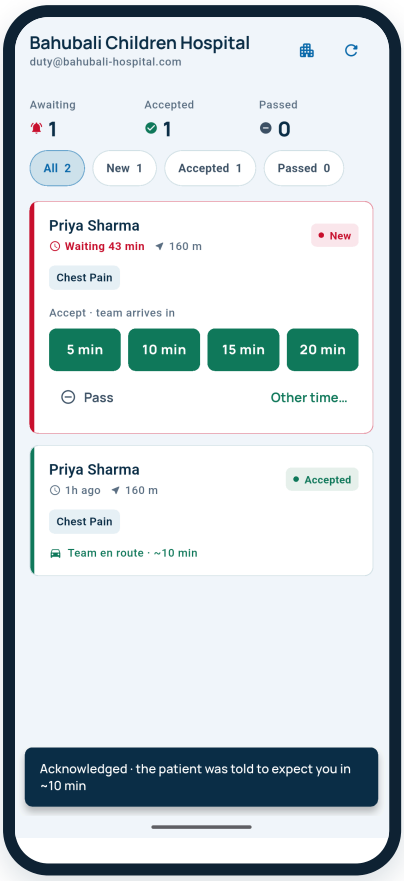
No refreshing needed

The patient's screen checks for news every few seconds and updates by itself, so they can keep their attention on the person who is unwell.

SECTION 3 · APP TOUR

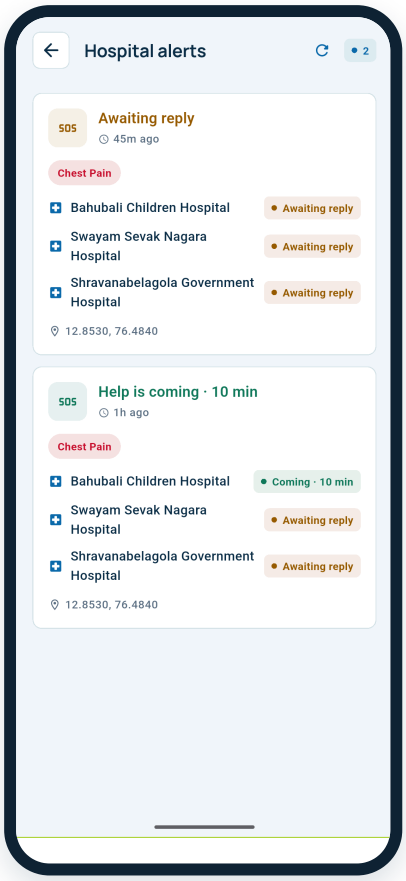
Hospital response, history and account

Both sides keep a record of what happened, and members stay in control of their account.



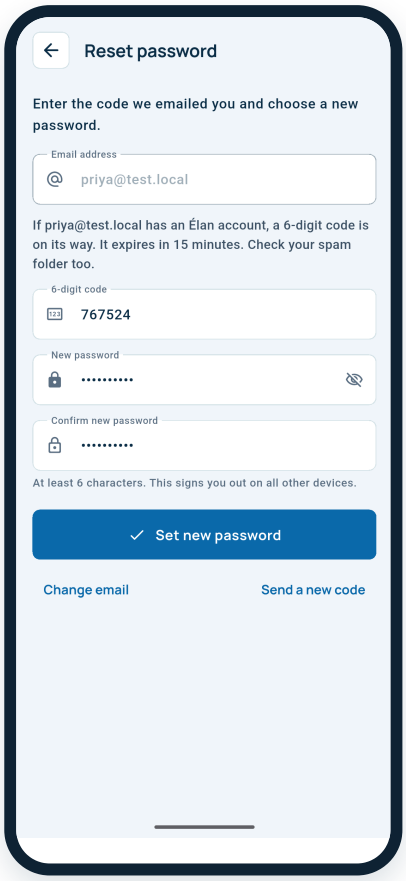
Hospital: accepted

The desk confirms the patient was told to expect the team in about 10 minutes, and the alert moves to "Accepted".



Patient: alert history

Every alert sent, with each hospital's reply: "Awaiting reply" or "Coming · 10 min".



Forgot password

A 6-digit code is emailed to the user. It works for 15 minutes, and setting a new password signs out all other devices.

One alert, several hospitals

All three hospitals see the same alert. The first to accept is shown to the patient as "coming"; the others can still respond or pass.

Locked out? Not for long

If a password is forgotten, the reset code arrives by email. For safety, the reply never reveals whether an email address has an account, and the code stops working after 15 minutes or 5 wrong tries.

SECTION 4

Safety by design

An emergency app must be honest about its limits and hard to misuse. These choices are built into Élan.

112

Never a replacement for 112

The first-launch notice, the Emergency screen and the alert status screen all point to 112, with a one-tap call button.

Hold to send, not tap

The SOS button must be held for 1.5 seconds. A pocket tap or a slip of the finger does not send an alert.

Real location only

If the phone cannot get a GPS fix, the app stops and says so. It never sends a guessed or default location.

Honest when offline

Without a connection, the app clearly says an alert was **not** sent, instead of pretending it was.

Approved hospital staff only

Staff can see patients' names, phone numbers and locations, so a new staff account stays locked until an administrator approves it.

Urgency is visible

On the hospital desk, unanswered alerts come first and their waiting timer turns red after 2 minutes.

Protected accounts

Passwords are stored scrambled (hashed) so nobody can read them, and repeated sign-in attempts are slowed down to stop guessing.

No payment in an emergency

Alerts are free, so nobody has to deal with a payment screen while in distress.

AN EXAMPLE OF ÉLAN IN USE

A 58-year-old man feels a crushing pain in his chest at home. His daughter calls 112, and while she waits she opens Élan, taps "Chest Pain" and holds the SOS button. Within seconds, three hospitals nearby see his location and symptoms. One of them accepts and answers "10 min". Her screen now reads "Help is on the way", with a countdown and a button to call that hospital directly. The hospital's team knows what to expect before they arrive. *(Illustrative example.)*

What Élan is not: it does not diagnose, treat or give medical advice, and it cannot guarantee that a hospital will respond or how quickly help will arrive. Hospitals decide whether they can take a patient.

SECTION 5

Your data and privacy

Élan collects only what it needs to get help to someone, and shares it only with the hospitals that person alerts.

WHAT	WHY IT IS NEEDED	WHO CAN SEE IT
Name and email	To create the account, sign in and reset a forgotten password.	The user. The name is shown to hospitals they alert.
Phone number (optional)	So a hospital can call the patient back.	Hospitals the user alerts.
Precise location	So hospitals know where to go. Read only on the Emergency screen.	Shared only when an alert is sent, and only with those hospitals.
Symptoms and message	So hospitals can prepare before arriving.	Hospitals the user alerts.
Health records (optional)	Patients, vital signs and reports the user chooses to keep.	Only the user.
Alert history	So the user can see what happened and who replied.	The user, and the hospitals involved.

Controls the user has

Delete account

From the Profile screen in the app, or from a web page on the Élan server. This removes the account together with its alerts and records.

Reset password

A code is sent by email. Changing the password signs the account out everywhere else, which helps if a phone is lost.

Read the policy

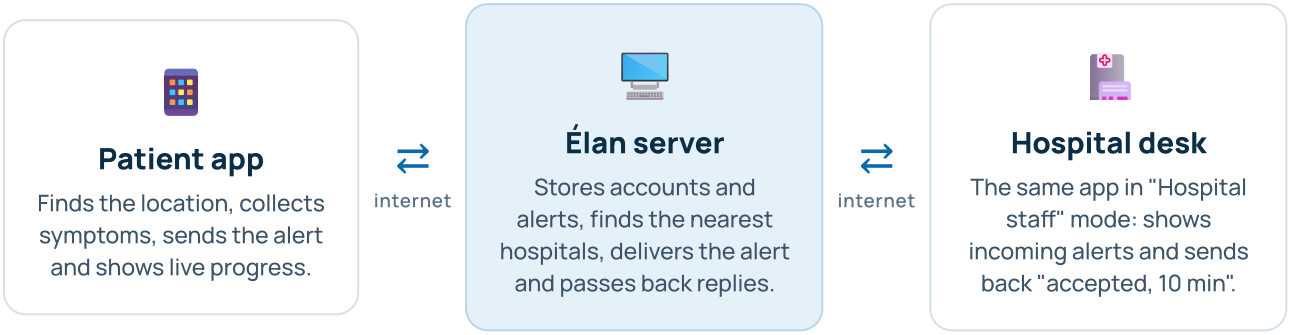
A full privacy policy is linked from the safety notice, the sign-in screen and the profile. It explains retention and the user's rights.

No advertising and no tracking or analytics tools are built into the app. Élan does not sell or share data for marketing.

SECTION 6

Behind the scenes

Élan has three parts that talk to each other over the internet. A simple way to picture it: the app is the *phone line*, the server is the *switchboard*, and the hospital desk is the *receiver*.



The server can also email hospitals the alert details with a map link, once email is set up. The patient's screen checks for updates every few seconds, so a hospital's reply appears without the patient doing anything.

For technical readers

PART	BUILT WITH
Mobile app	Flutter 3.38 (Dart), one code base for Android and iOS; custom light theme with the Manrope typeface
Server (backend)	Node.js with Express; REST API returning JSON
Database	SQLite (via sql.js), saved to a file on the server
Sign-in and security	JWT session tokens, bcrypt password hashing, rate limiting on sign-in and reset, input validation
Notifications	Email via Nodemailer (SMTP); Firebase Cloud Messaging push support
Payments	Razorpay support exists but is switched off; alerts are free in version 1
Phone features	GPS through the geolocator package; dialling and opening links through a small native Android/iOS bridge

SECTION 7

Testing, current status and next steps

How the app was tested

72
automated server tests (sign-in, alerts, hospital replies, deletion, password reset, free alerts) all passing

18
automated app tests (launch, safety notice, sign-in, hold-to-send, hospital desk) all passing

End to end
full alert journey run by hand on an Android emulator, from both the patient and hospital side

Done for version 1

- ✓ Hold-to-send SOS with live location and symptoms
- ✓ Live alert tracker with arrival countdown
- ✓ Hospital triage desk with one-tap accept
- ✓ Free alerts (no payment step)
- ✓ Safety notice and standing 112 button
- ✓ Privacy policy page
- ✓ Account deletion, in app and on the web
- ✓ Password reset by emailed code
- ✓ Staff approval before seeing patient data
- ✓ New brand, logo and launch screen

Before the Play Store launch

- ○ Host the server online with a secure (HTTPS) address
- ○ Connect an email service so reset codes and hospital emails are delivered
- ○ Set the official support email address
- ○ Fill in Google Play's Data safety and Health app forms (answers are already drafted)
- ○ Have the privacy policy and medical wording reviewed by a qualified person
- ○ Onboard real hospitals and approve their staff accounts

Ideas for the future

- → Live ambulance position and route on a map
- → More languages, starting with Hindi
- → Instant push notifications on the hospital desk

SECTION 8

Glossary of terms

Short explanations of words used in this report.

Alert	A request for help sent from the app to nearby hospitals, with location and symptoms.
112 / 108	India's national emergency number (112) and ambulance number (108). Élan always points users to these first.
GPS	The satellite system a phone uses to work out exactly where it is.
Server (backend)	A computer on the internet that stores accounts and alerts and passes messages between patients and hospitals.
Triage desk	The hospital-side screen where duty staff see incoming alerts, most urgent first, and decide how to respond.
ETA	"Estimated time of arrival": how many minutes until the hospital's team is expected to reach the patient.
Hold to send	The SOS button only sends after being held down for 1.5 seconds, which prevents accidental alerts.
HTTPS	A secure, encrypted way of sending data over the internet, so nobody in between can read it.
Hashing	Scrambling a password one way before storing it, so even the server's owners cannot read it.
Emulator	A virtual phone running on a computer, used to test the app.
Play Console	Google's website for publishing apps on the Google Play Store.
Data safety form	A form on the Play Console where developers declare what data an app collects and shares.



Élan — Emergency cardiac care

Report prepared by **Aditya Satwar** · 15 October 2026